FLORENT TARIOLLE

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Seeking a six-month research internship in world models · Available April 2027

Education

INSA Rouen Normandy & University of Rouen Normandy Expected 2027
Engineering degree, Computer Science and Artificial Intelligence (INSA; 2023–2027)
Master's degree, Machine Learning for Artificial Intelligence (University of Rouen; 2026–2027)

Lycée Dupuy de Lôme 2021 – 2023
Two-year intensive preparatory programme in mathematics and physics (CPGE MP)

Publications

DashVMC: Real-Time Discrete World Model Control in Geometry Dash Sep 2026
F. Tariolle and F. Yger. PTA workshop, NeurIPS 2026.
Paper · Demo · Code

- Developed a pixel-based world-model agent for Geometry Dash, refining a behaviour-cloned policy with PPO entirely in frozen-model rollouts.
- Trained offline on approximately two hours of gameplay; improved survival over cloning on three official levels and a held-out layout; **60-FPS control** on an RTX 2060.

Opportunistic Target Selection: Early Directional Commitment for Query-Efficient Black-Box Adversarial Attacks May 2026
F. Tariolle and F. Yger. CAp 2026 (Conférence sur l'Apprentissage automatique). Poster.
Paper · Code · Poster

- Developed a wrapper for score-based black-box adversarial attacks that uses early class-confidence signals to select a target, reducing class drift without gradient access.
- Evaluated 4,500 ImageNet attack runs across five classifiers. On ResNet-50, improved attack success by up to **27 percentage points** and reduced censored-mean attack iterations by 43%.

Research Projects and Awards

Finalist, Hack the World(s) Hackathon Jun 2026
Geometry-aware joint-embedding EEG · 24-hour JEPA and world-model challenge
Code · Presentation

- Selected as one of 100 participants from 650+ applicants for a hackathon sponsored by Yann LeCun. Our team finished among the top five of 25 teams.
- Built a model that learns from unlabeled EEG recordings; explored whether training that respects the curved geometry of covariance matrices leads to more structured representations.

Research and Industry Experience

InterDigital · Deep Learning Research Intern May – Sep 2026
Rennes, France

- Researched learned decoder-side filter selection as an alternative to neural in-loop filtering for VVC (H.266) video decoding, targeting compression and quantization artifacts.
- Contributed to **more than 10 patent applications** arising from this research.

InterDigital · Generative AI Consultant Oct 2026 – Present
Freelance, remote · Advising R&D teams on generative AI workflows, plugins, and skills.

- Continuing work on tools initiated during the research internship; guiding teams on AI-assisted development, reliability, and security.

Enedis · Machine Learning Intern Jan 2026 – Present
Part-time capstone, 25 hours/week · Rouen, France · Paused May–Sep 2026

- Developing a digital twin of the distribution network by mapping synthetic customer load profiles to distribution substations and linking these to primary substations.
- Combining DeepCourboGen-generated curves with measured substation loads; developing spatial and temporal adjustments for more realistic local profiles.

Open-Source Projects

VisualTorch · Maintainer and MCP Integration Lead Jul 2026 – Present
Official PyTorch Ecosystem project · 300+ GitHub stars

- Designed and implemented MCP integration to generate editable neural-network diagrams from PyTorch model definitions.
- Maintaining the architecture-visualization library and making model inspection available through agent tooling.

Rose · Co-founder and Lead Developer Sep – Dec 2025
Open-source League of Legends customization tool · 700+ GitHub stars · 20+ contributors

- Built core Python backend components, in-client JavaScript plugins, and their real-time WebSocket bridge; remain the project's top contributor.
- The project has since reached **30,000 daily active users** and 90,000 monthly active users.

Skills and Languages

Programming: Python, C++ **Frameworks:** PyTorch
Research: Self-supervised learning, world models, model-based RL and planning
Languages: French (native), English (TOEIC 970/990), Spanish (intermediate)

Interests

League of Legends (top 0.1% of ranked players; two LAN wins), bouldering, and strength training.

References

Florian Yger (INSA Rouen Normandy)
Research supervisor and co-author
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Frédéric Lefèbvre (InterDigital)
Internship and consulting supervisor
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